

The Gut Health Topic Summary Reports is a quick overview of gut-related genetics articles. Please see the linked articles for details and complete references.

[Article: Irritable Bowel Syndrome](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
SI	rs9290264	CA	A	0.29	Reduced sucrase-isomaltase enzyme
LCT	rs4988235	GG	G	0.49	GG = no lactase (lactose intolerant)
HTR3E	rs56109847	--	A	0.03	Big increase in IBS risk
HTR3A	rs1062613	CT	T	0.23	Increased risk of IBS-D
KLB	rs17618244	GG	G	0.82	GG only: faster transit time in IBS-D
TRPM8	rs10166942	TC	C	0.23	Slower colonic transit; Increased risk of IBS-C

[Article: Non-secretor](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
FUT2	rs601338	--	A	0.47	AA only: non-secretor of blood type, resistant to norovirus, low bifidobacteria; A/G: secretor of blood type
FUT2	rs1047781	AA	T	0.01	TT only: non-secretor if Japanese or Korean ancestry
FUT2	rs602662	GG	A	0.48	A/A: likely a non-secretor of blood type; A/G: secretor of blood type

[Article: FODMAPS and SI](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
SI	rs147207752	TT	C	0.001	FODMAPS diet less likely to work; low sucrase-isomaltase
SI	rs146785675	AA	G	0.005	FODMAPS diet less likely to work; low sucrase-isomaltase
SI	rs9290264	CA	A	0.29	FODMAPS diet less likely to work; low sucrase-isomaltase
SI	rs121912615	AA	C	0.002	Rare mutation; SI deficiency
SI	rs200451408	--	A	0.0001	Rare mutation; SI deficiency
SI	rs79717168	AA	C		Rare mutation; SI deficiency

[Article: Gut Microbiome](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
FUT2	rs601338	--	A	0.47	AA only: low/no Bifidobacteria
VDR	rs7974353	--	T	0.15	Influences Parabacteroides
LCT	rs182549	--	C	0.48	CC: No lactase as an adult; likely to have more bifidobacteria
SLC39A8	rs13107325	CC	T	0.07	Altered gut barrier function
APOA5	rs651821	TT	C	0.07	Reduced Bifidobacteria; higher triglycerides
ALDH1L1	rs2276731	TT	C	0.17	Higher SHA-98
IL4	rs2243250	CC	T	0.18	Increased C. difficile risk in IBD
TLR4	rs4986790	AA	G	0.05	increased risk of gram-negative bacterial infection, septic shock, and metabolic syndrome
TLR4	rs10759932	--	C	0.11	decreased risk of H. pylori infection

[Article: Mannose Binding Lectin](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
MBL2	rs1800450	CC	T	0.13	Decreased mannose binding lectin
MBL2	rs7096206	--	G	0.28	Decreased mannose binding lectin
MBL2	rs1800451	CC	T	0.03	Decreased mannose binding lectin
MBL2	rs5030737	GG	A	0.06	Decreased mannose binding lectin
MASP2	rs72550870	TT	C	0.02	Decreased mannose binding lectin

[Article: Aspirin & Colon Cancer](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
UGT1A6	rs1105879	AC	C	0.34	Reduced enzyme function; no colon cancer risk reduction for aspirin
SMAD7	rs4939827	TT	C	0.51	Decreased colon cancer risk with aspirin
TCFL2	rs7903146	CC	T	0.28	Decreased colon cancer risk with aspirin
CASC8	rs6983267	GG	T	0.47	Decreased colon cancer risk with aspirin
UGT1A6	rs2070959	AA	G	0.32	lower enzyme activity; no colon cancer risk reduction for aspirin

[Article: Meat and Colon Cancer](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
GATA3	rs1269486	--	A	0.24	Higher relative risk of colon cancer with increasing meat consumption
CCAT	rs6983267	GG	T	0.47	Decreased colon cancer, except in people who eat a lot of processed meats
GATA3	rs4143094	GG	T	0.25	20-40% higher relative risk of colon cancer with increasing meat consumption
AHR	rs2066853	GG	A	0.12	Increased risk of polyps with high meat intake
HAS2	rs9642855	--	C	0.36	increased risk of colon cancer in conjunction with a diet high in processed meat

[Article: IBD](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
NOD2	rs2066844	CC	T	0.04	2-35x increased risk for Crohn's disease
NOD2	rs2066845	--	C	0.01	2-35x increased risk for Crohn's disease
NOD2	rs2066842	CT	T	0.24	Increased risk of IBD
NOD2	rs2066843	--	T	0.24	Increased risk of IBD
TLR4	rs4986790	AA	G	0.05	Increased risk of IBD
IBD5	rs6596075	--	G	0.17	Lower risk of IBD
ETS2	rs2836882	AG	A	0.27	Lower risk of IBD
PTPN2	rs1893217	AA	G	0.15	increased relative risk of IBD, increased risk of invasive E. coli in IBD patients

[Article: Emulsifiers and Leaky Gut](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
IL10	rs1800871	AG	G	0.72	increase inflammation, increase risk of inflammatory bowel disease
IL10	rs3024505	GG	A	0.14	increase inflammation, increase risk of inflammatory bowel disease
IL10	rs1800896	TT	C	0.45	increase inflammation, increase risk of inflammatory bowel disease
NOD2	rs2066844	CC	T	0.04	Increased risk of Crohn's disease; increased intestinal permeability
NOD2	rs2066845	--	C	0.01	Increased risk of Crohn's disease
NOD2	rs2066847	--	I	0.91	Increased risk of Crohn's disease

[Article: Gut Mucosa](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
MUC2	rs11825977	GG	A	0.19	decreased relative risk of inflammatory bowel diseases; decreased risk of Parkinson's; likely higher MUC2 expression (interpretation)
MUC2	rs2856111	CC	C	0.14	slightly increased relative risk of IBD; likely lower MUC2 expression (interpretation)
MUC2	rs7396030	--	G	0.77	G/G: increased MUC2, increased risk of gallstones
MUC5AC	rs868903	--	T	0.47	increased relative risk of stomach cancer
MUC5AC	rs16866419	AA	C	0.04	decreased MUC5AC production (GWAS study)
MUC5AC	rs75401036	GG	A	0.01	decreased MUC5AC production (GWAS study)
MUC5AC	rs2688513	--	G	0.17	higher mortality rate in colorectal cancer
CLCA1	rs2734705	--	G	0.84	G/G: likely more mucin secretion; increased risk of intestinal ileus (lack of motility) in infants with cystic fibrosis

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purposes only. It is not intended to diagnose or treat any disease. Please visit www.GeneticLifehacks.com for more detailed information and full references.